



Cambridge (CIE) IGCSE Biology



Your notes

Human Impact: Biodiversity, Pollution & Conservation

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Ensuring Food Supply



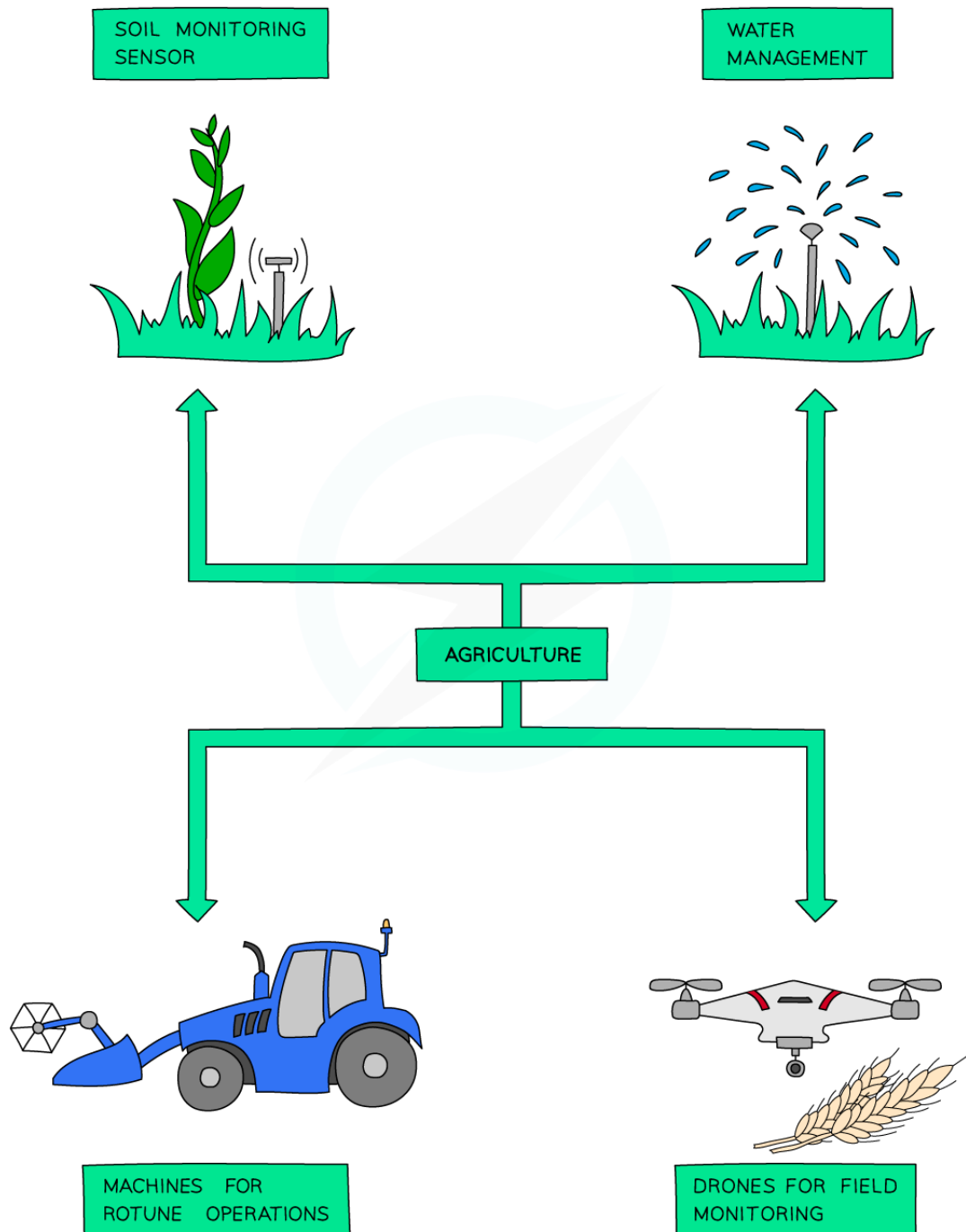
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Intensive Food Production

- Making food production more **intensive** means producing food more **efficiently** with a finite amount of land and other resources
- Modern technology has increased food supply substantially in the following ways:
 - **Agricultural machinery** has replaced humans and **improved efficiency** due to the ability to farm **much larger areas** of land
 - **Chemical fertilisers improve yields** – fertilisers increase the amount of nutrients in the soil for plants, meaning that they can grow larger and produce more fruit
 - **Insecticides and herbicides** – these chemicals kill off unwanted insects and weed species, meaning that there is **less damage done** to plants and fruit lost to insects (insecticides), as well as **reducing competition** from other plant species (herbicides)
 - **Selective breeding** – animals and crop plants which produce a large yield are selectively bred to produce breeds that **reliably produce high yields**



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Modern agricultural processes allows for cultivation of much larger areas of land for crop plants



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Intensive Livestock Farming

- In developed countries, **large numbers of livestock** are often kept in an area that would **not normally be able to support more than a very small number**
- They are often fed **high energy foods**, regularly given medication such as **antibiotics** as a preventative measure against disease and kept in **artificially warm temperatures** and small spaces that **do not allow for much movement**

Advantages of intensive livestock farming

- Less land is required to produce large amounts of food
- Food can be produced all year round in controlled environments
- The cost of production is lower
 - therefore food can be sold at cheaper prices
- There is a lower requirement for labour to produce the food

Disadvantages of intensive livestock farming

- Intensive farming techniques tend to use
 - herbicides which may result in **eutrophication**
 - pesticides which can cause **negative effects** on the foodchain
- Ethical issues associated with cruel treatment of animals in intensive farms
- Natural habitats may be destroyed to maximise land available
- Ecological issues with intensive farming include:
 - **reduction in biodiversity** in areas where large amounts of land are used to graze cattle (as only grass is grown so in effect it becomes a monoculture)
 - overgrazing can lead to **soil erosion**
 - large numbers of cattle produce large amounts of **methane**, a greenhouse gas

Monocultures

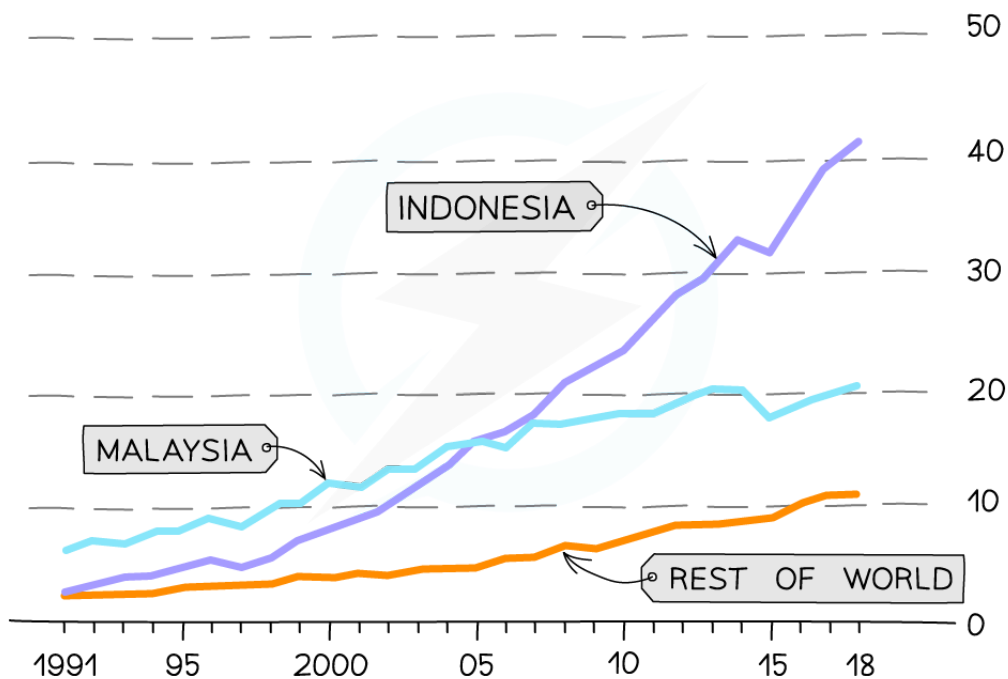
- Monoculture farming means that on a given area of agricultural land **only one type of crop is grown** (eg trees for palm oil grown in Indonesian rainforest)Th



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- is large scale growth of a single variety of plant **does not happen naturally** in ecosystems, where there are **usually many different species of plants** growing which, in turn, **support many species of animals** (high biodiversity)
- In monocultures, **biodiversity is much lower**
- Another issue with monocultures is the **increase in pest populations** – if a particular pest feeds on a crop, farming it in large areas repeatedly means there is an ample supply of food for the pest, causing the population to increase
- Often farmers will spray **insecticides** onto crops in order to control the pests. This leads to:
 - harmless insects being killed as well
 - pollution by pesticides (which are often **persistent chemicals** which accumulate in food chains)
 - in many instances where they are used repeatedly for specific pests, the pests may eventually become **resistant** to them, reducing their effectiveness

PALM-OIL PRODUCTION, TONNES, m



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Palm oil production has increased rapidly over the last 30 years



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The Importance of Biodiversity

Human Pressures on Other Species

Biodiversity

- Is defined as the **number of different species** that live in a particular area
- Human activities have tended to **force biodiversity downwards**, whereas, high biodiversity is needed for stable ecosystems
- **Habitat destruction** by humans is a major downward pressure on biodiversity

Reasons for Habitat Destruction

- The increasing human population of the planet is causing destruction of many habitats from rainforest to woodland to marine
- Many habitats are destroyed by humans to make space for other economic activities, or by pollution from these activities, and this reduces the biodiversity of these areas
- This interrupts food chains and webs, meaning that more species may die because their prey is gone
- The main reasons for habitat destruction include:



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REASON	EXPLANATION
CLEARING LAND FOR FARMING AND HOUSING	– CROPS, LIVESTOCK AND HOMES ALL TAKE UP A LARGE AMOUNT OF SPACE – AS THERE IS AN INCREASING POPULATION AND DEMAND FOR FOOD, THE AMOUNT OF LAND AVAILABLE FOR THESE THINGS MUST BE INCREASED BY CLEARING HABITATS SUCH AS FORESTS (DEFORESTATION)
EXTRACTION OF NATURAL RESOURCES	– NATURAL RESOURCES SUCH AS WOOD, STONE AND METALS MUST BE GATHERED TO MAKE DIFFERENT PRODUCTS. – THEREFORE MANY TREES ARE CUT DOWN, DESTROYING FOREST HABITATS. IN ADDITION, SOME RESOURCE EXTRACTION TAKES UP A LARGE AMOUNT OF SPACE – FOR EXAMPLE: MINING, WHICH MEANS THAT THE LAND MUST BE CLEARED FIRST
MARINE POLLUTION	– HUMAN ACTIVITIES LEAD TO THE POLLUTION OF MARINE HABITATS – IN MANY PLACES, OIL SPILLS AND OTHER WASTE POLLUTES THE OCEANS, KILLING SEA LIFE – IN ADDITION, EUTROPHICATION CAN OCCUR WHEN FERTILISERS FROM INTENSIVELY FARMED FIELDS ENTERS WATERWAYS – THIS CAUSES A HUGE DECREASE IN BIODIVERSITY IN THESE AREAS AS MOST AQUATIC SPECIES LIVING IN THESE WATERWAYS DIE FROM LACK OF OXYGEN

Deforestation

- Deforestation is the **clearing of trees** (usually on a large scale)
- If trees are replaced by replanting it can be a **sustainable** practise
- Generally the trees are being cleared for the **land to be used in a different way** (for building, grazing for cattle, planting of monocultures such as palm oil plantations etc) and therefore it is not sustainable
- As the amount of the Earth's surface covered by trees decreases, it causes increasingly negative effects on the environment and is a **particularly severe example of habitat destruction**
- Undesirable effects of deforestation include:

- **Extinction** of species
- Loss of **soil**
- **Flooding**
- Increase of **carbon dioxide** in the atmosphere

EFFECT	CONSEQUENCE
EXTINCTION / LOSS OF BIODIVERSITY	– FOREST HABITATS, ESPECIALLY TROPICAL RAINFORESTS, HAVE A HUGE RANGE OF BIODIVERSITY AND AS HABITAT IS DESTROYED IT CAUSES THE LOSS OF LARGE NUMBERS OF PLANT AND ANIMAL SPECIES – MANY OF THESE SPECIES ARE ONLY FOUND IN THESE AREAS AND THEREFORE WILL BECOME EXTINCT
SOIL EROSION	– TREE ROOTS HELP TO STABILISE THE SOIL, PREVENTING IT FROM BEING ERODED BY RAIN – TREES WILL USUALLY TAKE UP NUTRIENTS AND MINERALS FROM THE SOIL THROUGH THEIR ROOTS – WITHOUT TREES, NUTRIENTS AND MINERALS WILL REMAIN UNUSED IN THE SOIL SO WILL BE WASHED AWAY INTO RIVERS AND LAKES BY RAIN (LEACHING) – THIS LOSS OF SOIL NUTRIENTS IS PERMANENT AND MAKES IT VERY DIFFICULT FOR FOREST TREES TO REGROW, EVEN IF THE LAND IS NOT CULTIVATED WITH CROP PLANTS OR GRASS FOR CATTLE
FLOODING	– WITHOUT TREES THE TOPSOIL WILL BE LOOSE AND UNSTABLE SO WILL BE EASILY WASHED AWAY BY RAIN, INCREASING THE RISK OF FLASH FLOODING AND LANDSLIDES
INCREASED CARBON DIOXIDE IN ATMOSPHERE	– TREES CARRY OUT PHOTOSYNTHESIS DURING WHICH THEY TAKE IN CARBON DIOXIDE AND RELEASE OXYGEN – THE REMOVAL OF SIGNIFICANT NUMBERS OF TREES MEANS LESS CARBON DIOXIDE IS BEING REMOVED FROM THE ATMOSPHERE (AND LESS OXYGEN RELEASED) – WHEN AREAS OF LAND IN FORESTS ARE CLEARED FOR LAND USE, THE TREES ARE OFTEN BURNED AS OPPOSED TO BEING CUT DOWN. THIS RELEASES CARBON DIOXIDE (IT IS AN EXAMPLE OF COMBUSTION), FURTHER INCREASING CARBON DIOXIDE LEVELS IN THE ATMOSPHERE AND CONTRIBUTING TO GLOBAL WARMING

Water Pollution



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Untreated Sewage & Excess Fertiliser

- Human activities have led to the pollution of land, water and air
- Pollution comes from a variety of sources, including industry and manufacturing processes, waste and discarded rubbish, chemicals from farming practices, nuclear fall-out, and untreated sewage



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POLLUTANT	SOURCE / CAUSE	EFFECT
UNTREATED SEWAGE	LACK OF SEWAGE TREATMENT PLANTS IN INHABITED AREAS (DUE TO POOR INFRASTRUCTURE / LACK OF MONEY) MEANING SEWAGE RUNS / IS PUMPED INTO STREAMS OR RIVERS.	PROVIDES A GOOD SOURCE OF FOOD FOR BACTERIA WHICH INCREASE RAPIDLY, DEPLETING THE OXYGEN DISSOLVED IN THE WATER (AS THEY RESPIRE AEROBICALLY) AND CAUSING DEATH OF AQUATIC ORGANISMS SUCH AS FISH – KNOWN AS EUTROPHICATION
CHEMICAL WASTE	CHEMICALS SUCH AS HEAVY METALS LIKE MERCURY CAN BE RELEASED FROM FACTORIES INTO RIVERS AND OCEANS OR LEACH INTO LAND SURROUNDING THE FACTORIES	MANY HEAVY METALS AND OTHER CHEMICALS ARE PERSISTENT – THEY DO NOT BREAK DOWN AND SO CAN BUILD UP IN FOOD CHAINS (KNOWN AS BIOACCUMULATION), POISONING THE TOP CARNIVORES
DISCARDED RUBBISH	MUCH RUBBISH CONSISTS OF PLASTIC THAT IS EITHER DISCARDED OR BURIED IN LANDFILLS.	MUCH RUBBISH, SUCH AS THAT MADE FROM PLASTIC, IS NON-BIODEGRADABLE AND REMAINS IN THE ENVIRONMENT FOR HUNDREDS OF YEARS. ANIMALS ALSO EAT THE PLASTIC AS IT BREAKS INTO SMALLER PIECES (ESPECIALLY IN THE OCEAN) AND IT CAN GET INTO FOOD CHAINS THIS WAY
FERTILISERS	RUNOFF FROM AGRICULTURAL LAND IF APPLIED IN TOO HIGH A CONCENTRATION	CAUSES ALGAL BLOOMS WHICH THEN DIE AND PROVIDE A GOOD SOURCE OF FOOD FOR DECOMPOSING BACTERIA WHICH INCREASE RAPIDLY, DEPLETING THE OXYGEN DISSOLVED IN THE WATER (AS THEY RESPIRE AEROBICALLY) AND CAUSING DEATH OF AQUATIC ORGANISMS SUCH AS FISH KNOWN AS EUTROPHICATION
INSECTICIDES & HERBICIDES	SPRAYED ON CROPS TO PREVENT DAMAGE BY INSECTS AND GROWTH OF WEEDS	BIOACCUMULATION, LOSS OF BIODIVERSITY, DAMAGE TO BENEFICIAL INSECTS, CAN BUILD UP IN SOIL TO TOXIC CONCENTRATIONS AND HARM OTHER ORGANISMS
NUCLEAR FALLOUT	RADIOACTIVE PARTICLES THAT GET INTO THE ENVIRONMENT FROM ACCIDENTAL LEAKAGE FROM NUCLEAR POWER PLANTS OR EXPLOSION OF A NUCLEAR BOMB	SOME RADIOACTIVE PARTICLES HAVE LONG HALF-LIVES AND CAN REMAIN IN THE ENVIRONMENT FOR MANY YEARS. THEY CAN CAUSE INCREASED RISKS OF CANCER AND SMALLER PARTICLES CAN BE CARRIED BY WINDS HUNDREDS OF MILES FROM THE ORIGINAL SITE OF EXPOSURE



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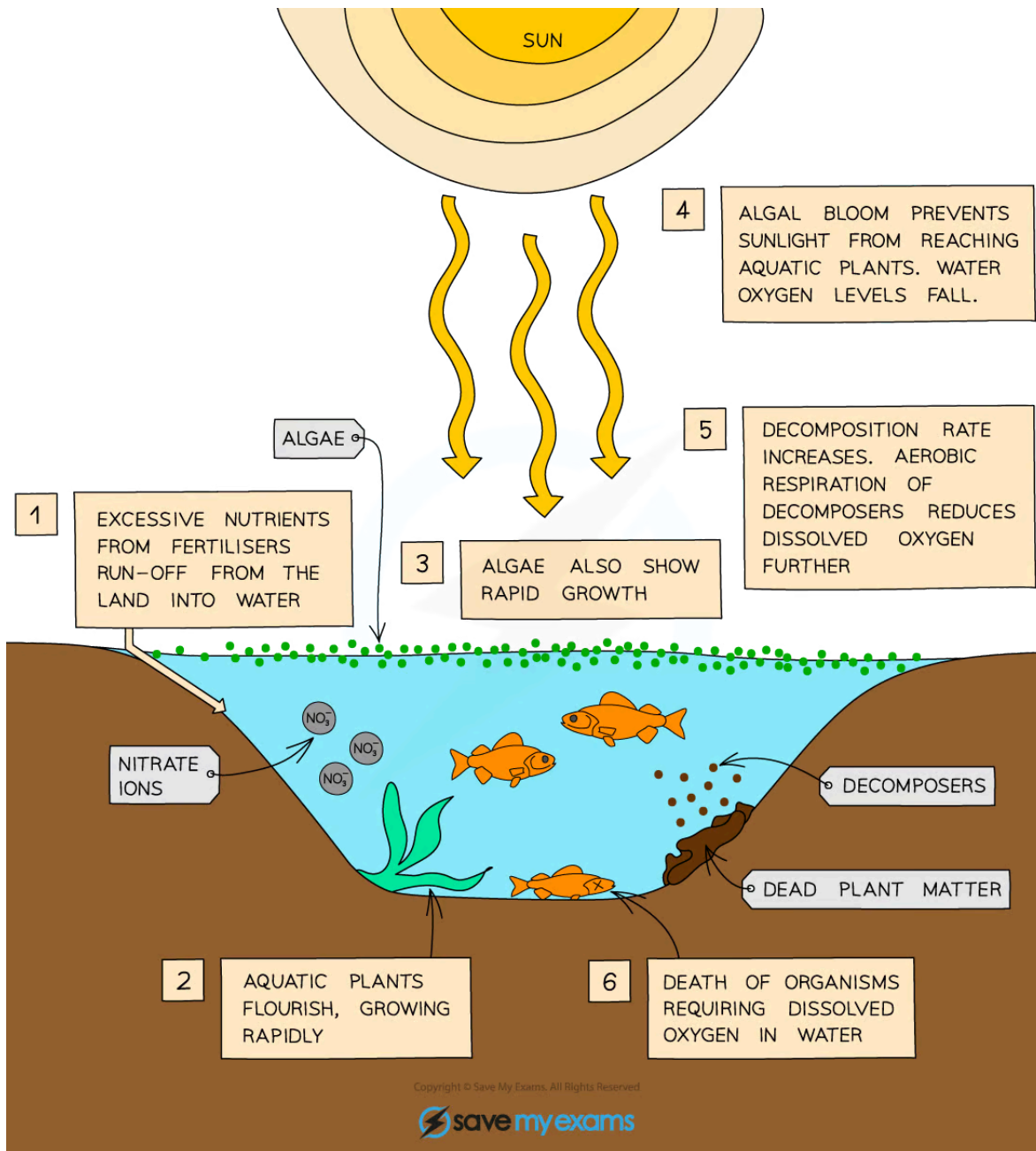
METHANE	CATTLE FARMING, RICE FIELDS, LANDFILLS	METHANE IS A GREENHOUSE GAS WHICH CONTRIBUTES TO THE ENHANCED GREENHOUSE EFFECT THAT IS CAUSING CLIMATE CHANGE
CARBON DIOXIDE	PRODUCED WHEN FOSSIL FUELS ARE BURNT, ALSO RELEASED WHEN TREES ARE BURNT TO CLEAR LAND FOR HUMAN USE	CARBON DIOXIDE IS A GREENHOUSE GAS WHICH CONTRIBUTES TO THE ENHANCED GREENHOUSE EFFECT THAT IS CAUSING CLIMATE CHANGE

Eutrophication: Extended

- Runoff of fertiliser from farmland enters the water and causes **increased growth of algae** and water plants
- The resulting 'algal bloom' **blocks sunlight** so water plants on the bottom start to die, as does the algae when competition for nutrients becomes too intense
- As water plants and algae die in greater numbers, **decomposing bacteria increase in number** and **use up the dissolved oxygen** whilst respiring aerobically
- As a result there is less oxygen dissolved in water, so **aquatic organisms such as fish and insects may be unable to breathe** and would die



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Sequence of events causing eutrophication in lakes and rivers



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Other Pollution

Plastic Pollution

- Plastics have a large negative impact on both land and water habitats due to their non-biodegradability
- In marine habitats:
 - Animals often try to **eat plastic** or **become caught in it**, leading to injuries and death
 - As the plastic breaks down it can **release toxins** that affect marine organisms
 - Once it has broken down into **very small particles**, it is commonly ingested by animals and **enters the food chain**

On land:

- Plastic is generally disposed of by **burying in landfills**
- As it breaks down, it releases **toxins** into the surrounding soil and as such the **land is no good for growing crops or grazing animals** and can only be used for building on several decades after burial

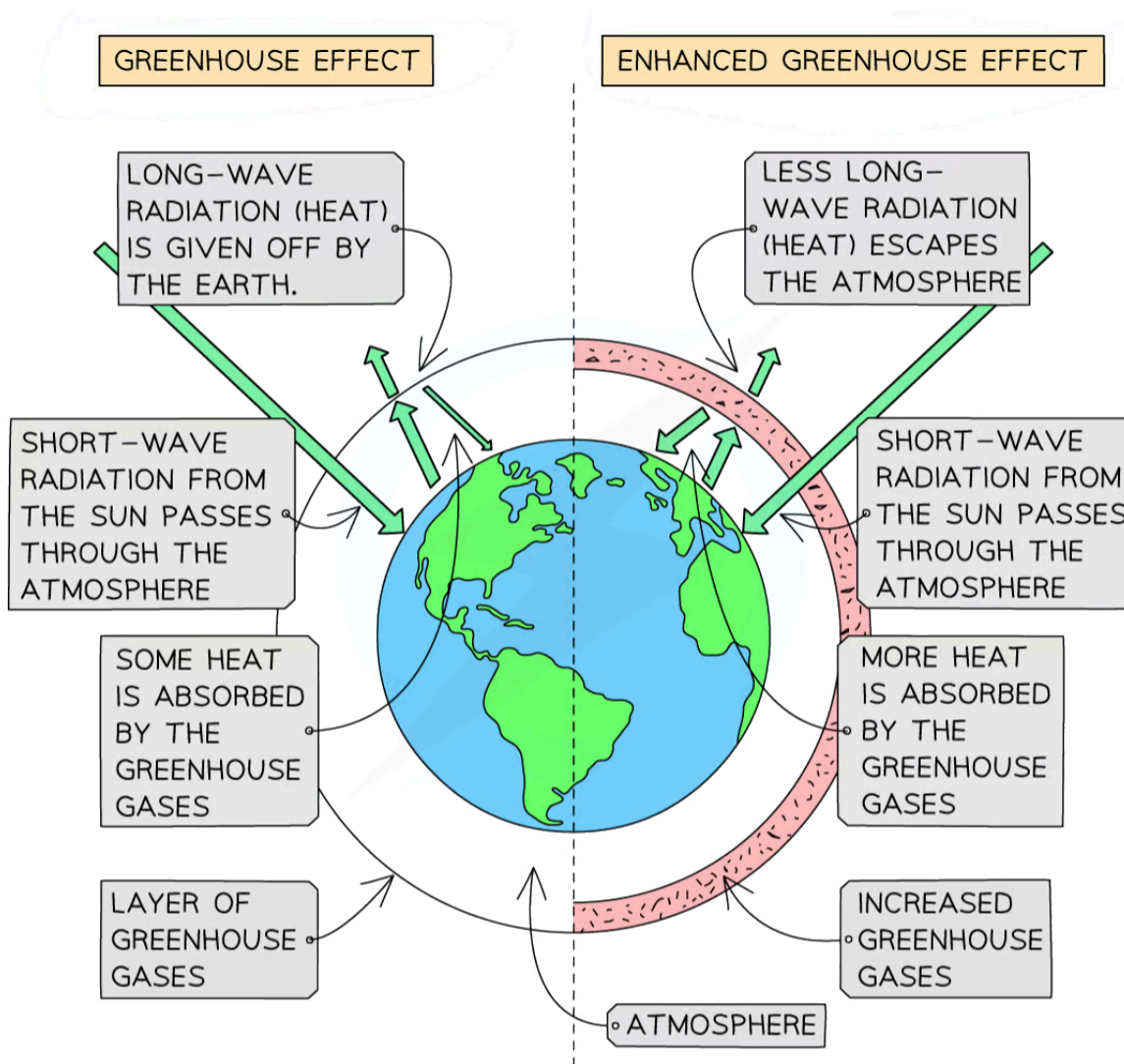
Air Pollution

Methane and Carbon Dioxide

- Both gases insulate the Earth and act as a 'blanket' around the atmosphere
- Higher levels of both have led to global warming and climate change
- Human activity has increased levels of both gases in the atmosphere
 - Burning fossil fuels increases carbon dioxide
 - Keeping livestock generates methane gas
 - Global warming melts the permafrost in sub-polar regions, which results in even more trapped methane being released into the atmosphere



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Sustainability

Sustainable Use of Resources

- We use many resources from the Earth; some, such as food, water and wood, are **sustainable resources**
- A sustainable resource is one which is **produced as rapidly as it is removed from the environment so that it does not run out**
- Some resources, such as **fossil fuels** (coal, oil and natural gas), are **non-renewable** because what we use cannot be replaced
- These resources, once used, cannot be produced anymore and so they need to be **conserved** by **reducing the amount we use** and finding other, **sustainable resources to replace them**
- Fossil fuels are being used as an **energy source** in increasing amounts
- In addition, they are the **raw materials** for many other products we make – eg almost all **plastics** that are made start with **oil** as a raw material
- Some products, especially those made from **paper, plastic, glass or metal**, can be **reused and recycled** – this reduces waste in the environment and reduces the amounts of raw materials and energy needed to make new products
- Some resources, such as **forests and fish stocks**, can be maintained – enabling us to **harvest them sustainably** so that they will **not run out in the future**

Sustainable Production Examples: Extended

- Sustainable development is defined as **development providing for the needs of an increasing human population without harming the environment**

When developing the way in which we use resources to manage them sustainably, we have to **balance conflicting demands** – eg:

- the need for **local people** to be able to utilise the resources they have in their immediate environment with the needs of **large companies** to make money from resources such as forests and fish
- the need for balancing the **needs of humans for resources** with the **needs of the animals and plants** that live in the areas the resources are taken from (preventing loss of habitat and extinction)
- the need to balance what **current populations need** with what **future populations might need** – for example if we harvest all the fish we need to feed people now, this might lead to overfishing which would deplete stocks for future generations

For development to occur sustainably, **people need to cooperate at local, national and international levels** in the planning and management of resources

Sustaining Forests

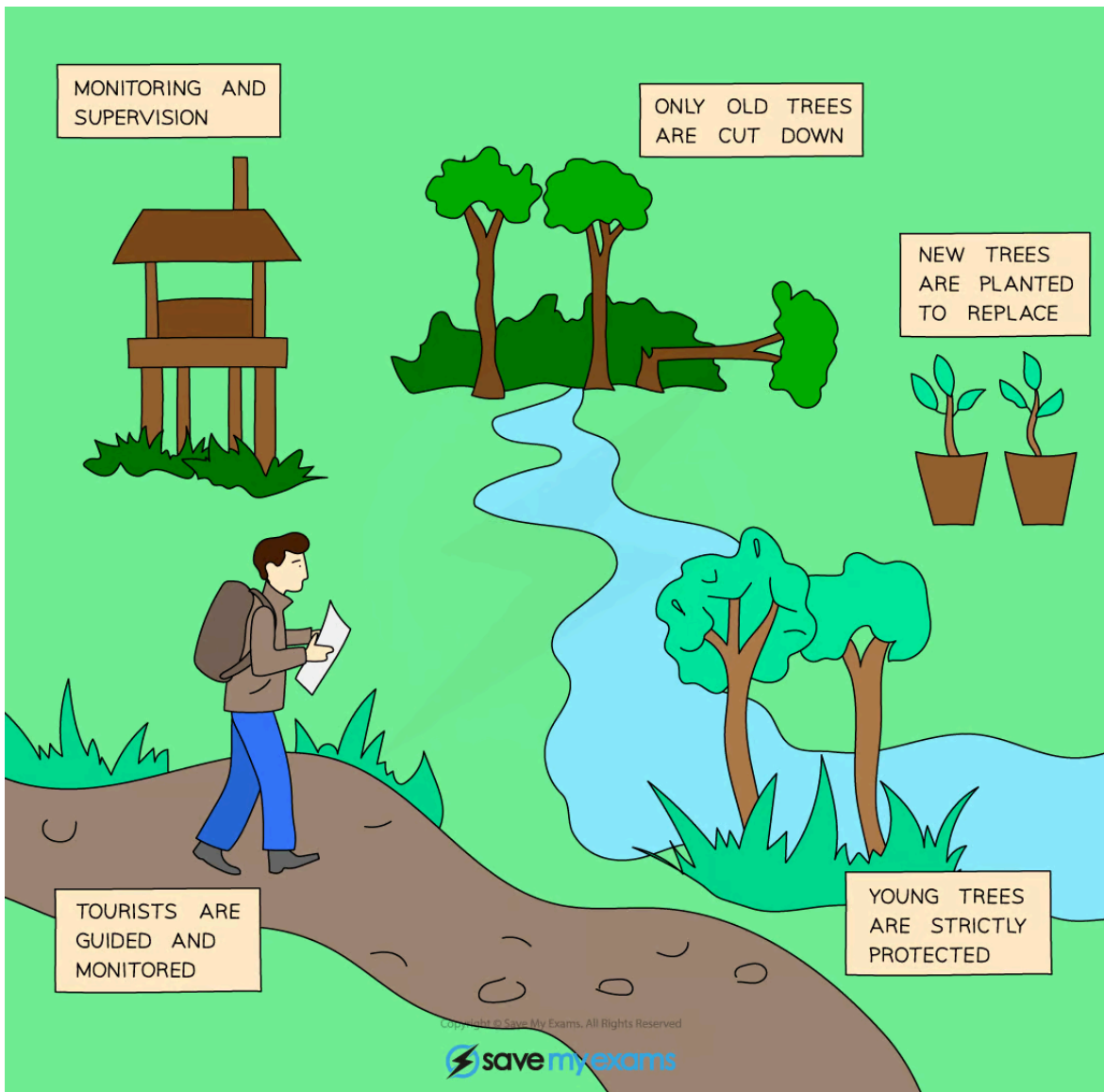
- Forests are needed to produce **paper products** and **provide wood for timber**
- Much of the world's paper is now produced from forests which **replant similar trees when mature trees are cut**, ensuring that there will be adequate supply in the future
- Tropical hardwoods such as teak and mahogany take many years to regrow but are highly desirable for furniture
- Using these types of wood has now been made more sustainable due to the **introduction of several schemes** designed to **monitor logging companies** and track the wood produced (eg the Forestry Stewardship Council)
- **Education** helps to ensure logging companies are aware of sustainable practices and consumers are aware of the importance of buying products made from sustainable sources



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More efforts are being made to manage forests sustainably so consumers know they are not causing damage to forests

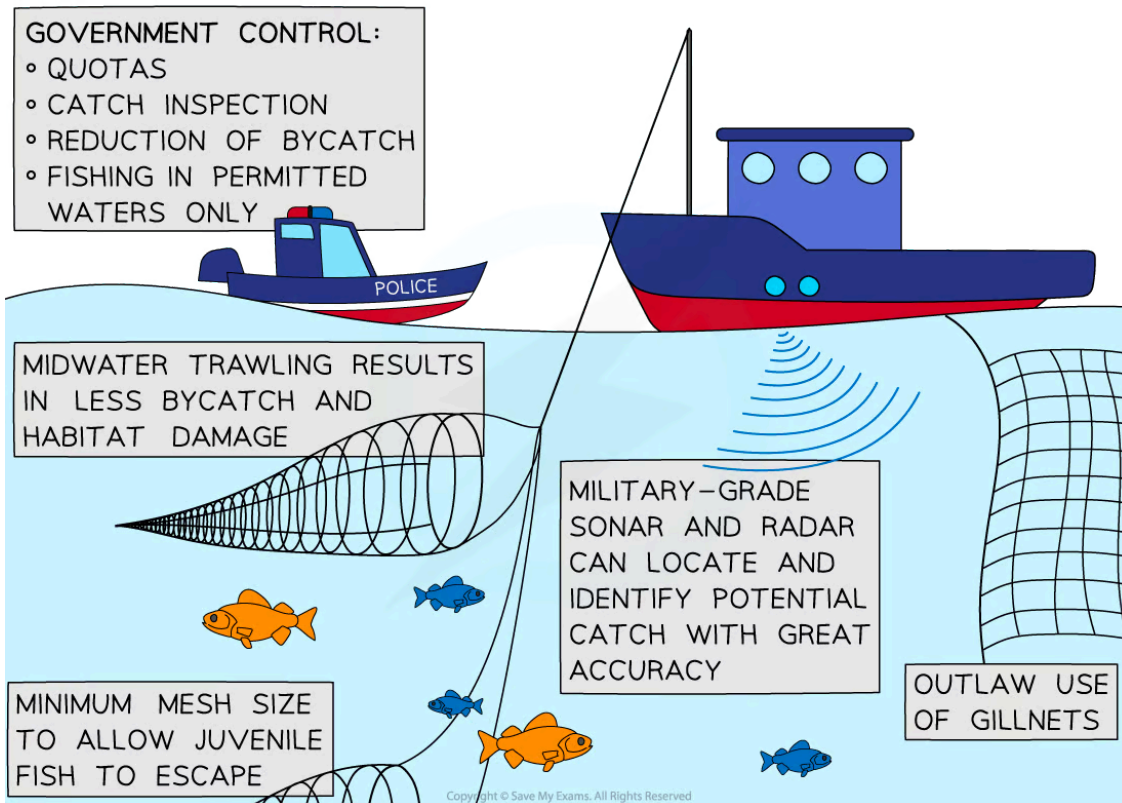
Sustaining Fish Stocks

- Managing fish stocks sustainably includes:
 - Controlling the **number of fish** caught each year (quotas)
 - Controlling the **size of fish** caught (to ensure there are enough fish of a suitable age for breeding remaining)



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- Controlling the **time of year** that certain fish can be caught (to prevent large scale depletion of stocks when fish come together in large numbers in certain areas to breed)
- **Restocking** (breeding and keeping offspring until they are large enough to survive in their natural habitat then releasing)
- **Educating fishermen** as to local and international laws and **consumers** so they are aware of types of fish which are not produced sustainably and can avoid them when buying fish



Ways of controlling fishing to make it more sustainable



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Endangered Species

Conservation for Endangered Species

Endangered Species

- An endangered species is at risk of becoming **extinct**
- There are several reasons why a species can become endangered – the **population of the species may fall below a critical level** due to
 - **hunting**
 - **climate change**
 - **pollution**
 - **loss of habitat**
 - **introduction of non-native species** that outcompete native species
- Endangered species can be helped by **conservation measures** such as:
 - **education** programmes
 - **captive breeding** programmes
 - **monitoring and legal protection** of the species and of their habitats
 - **seed banks** as a conservation measure for plants – seeds of endangered plant species are carefully stored so that new plants may be grown in the future
- A species may be at risk of becoming extinct **if there is not enough genetic variation in the population** as random changes in the environment may quickly cause extinction because the **remaining organisms are all very similar and may not have the adaptations to survive such changes**
- There are moral, cultural and scientific reasons for conservation programmes, including:
 - **reducing extinction rates** of both plant and animal species
 - keeping damage to food chains and food webs to a minimum and **protecting vulnerable ecosystems** (eg the rainforests)
 - **protecting our future food supply** and **maintaining nutrient cycles and possible sources of future medical drugs and fuels**



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Reasons for Conservation

Reasons for Conservation: Extended

Extended Tier Only

- There are numerous reasons why **conservation programmes are important**
 - Maintaining or increasing biodiversity
 - Which allows ecosystems to remain stable
 - Reducing extinction
 - Helps to retain iconic species and maintain biodiversity
 - Protecting vulnerable ecosystems which would have been quickly lost to human activity
 - Maintaining ecosystem functions
 - Nutrient cycling eg. carbon cycling to hold back climate change
 - Resource provision, such as
 - Food - making sure we have enough for the population
 - Drugs - having access to plants for plant-based remedies
 - Fuel - for important activities such as cooking
 - Genes - so the gene pool remains wide and variety exists in all species

Conservation Techniques: Extended

Extended Tier Only

- Certain conservation techniques can be used to **maintain biodiversity**
- Examples include
 - **Artificial insemination (AI)** in captive breeding programmes
 - This allows large numbers of offspring to be produced without the need for conventional sexual intercourse between males and females
 - **In vitro fertilisation (IVF)** in captive breeding programmes
 - This allows gametes with **known alleles** to be used in ensuring the next generation remains biodiverse

Risks to a species

- If its population size decreases, a species will experience **reduced genetic variation**
- This renders the species **more susceptible** to environmental change
- The species is **less resilient** and has a greater **risk of extinction**



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